Date: DD MM YY

CPU Scheduling Algorithms

Formulas :

1. Turn Around Time : TAT = CT − AT
2. Waiting Time : WT = TAT − BT
3. Average TAT = (sum of all TAT) ÷ n
4. Average WT = (sum of all WT) ÷ n

Note: CT is read from the Gantt chart (the time at which the process finishes).

Terms :

AT = Arrival Time (when the process enters the ready queue)
BT = Burst Time (CPU time the process needs)
CT = Completion Time (when the process finishes)
TAT = Turn Around Time (total time from arrival to finish)
WT = Waiting Time (time spent waiting in the ready queue)
TQ = Time Quantum (used only in Round Robin)
n = total number of processes

Types :

Non-preemptive : CPU is not taken back until the process finishes. Examples: FCFS, Priority (non-preemptive).

Preemptive : CPU can be taken from a running process. Examples: Round Robin (RR), SJF preemptive (SRTF).

Steps for every question :

1. Draw the Gantt chart (start from 0, write Idle if CPU is empty)
2. Read CT of every process from the Gantt chart
3. Find TAT = CT − AT
4. Find WT = TAT − BT
5. Take the averages

Date: DD MM YY

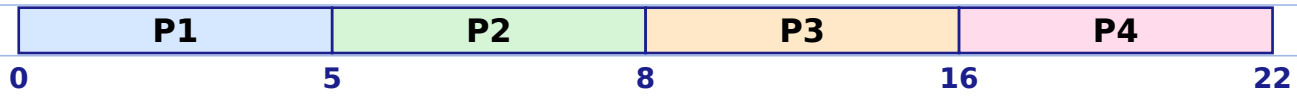
1. FCFS (First Come First Serve) — Non-preemptive

Rule : Processes run in the order in which they arrive.

1 Given Data :

Process	AT	BT
P1	0	5
P2	1	3
P3	2	8
P4	3	6

2 Gantt Chart :



3 Calculation :

CT = end time in the Gantt chart

Process	AT	BT	CT	TAT = CT − AT	WT = TAT − BT
P1	0	5	5	5 − 0 = 5	5 − 5 = 0
P2	1	3	8	8 − 1 = 7	7 − 3 = 4
P3	2	8	16	16 − 2 = 14	14 − 8 = 6
P4	3	6	22	22 − 3 = 19	19 − 6 = 13

4 Averages :

Avg WT = (0 + 4 + 6 + 13) ÷ 4 = 23 ÷ 4 = 5.75

Avg TAT = (5 + 7 + 14 + 19) ÷ 4 = 45 ÷ 4 = 11.25

Final Answer : | **Avg WT = 5.75** | **Avg TAT = 11.25**

Date: DD MM YY

2. Round Robin (RR) — Preemptive

Rule : Every process gets a fixed Time Quantum (TQ).

If work remains, the process goes to the back of the ready queue.

Note: If a new process arrives at the same time a process is preempted, the new process enters the queue first.

Example A : TQ = 3

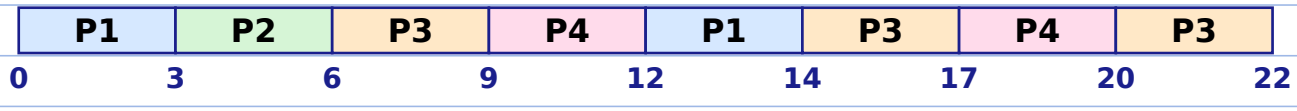
1 Given Data :

Process	AT	BT
P1	0	5
P2	1	3
P3	2	8
P4	3	6

2 Ready Queue :

Time	Process	Remaining BT	Queue after this slot
0 – 3	P1	5 − 3 = 2	P2, P3, P4, P1
3 – 6	P2	3 − 3 = 0 ✔	P3, P4, P1
6 – 9	P3	8 − 3 = 5	P4, P1, P3
9 – 12	P4	6 − 3 = 3	P1, P3, P4
12 – 14	P1	2 − 2 = 0 ✔	P3, P4
14 – 17	P3	5 − 3 = 2	P4, P3
17 – 20	P4	3 − 3 = 0 ✔	P3
20 – 22	P3	2 − 2 = 0 ✔	–

3 Gantt Chart :



4 Calculation : CT = last end time of the process

Process	AT	BT	CT	TAT = CT − AT	WT = TAT − BT
P1	0	5	14	14 − 0 = 14	14 − 5 = 9
P2	1	3	6	6 − 1 = 5	5 − 3 = 2
P3	2	8	22	22 − 2 = 20	20 − 8 = 12
P4	3	6	20	20 − 3 = 17	17 − 6 = 11

Date: DD MM YY

2. Round Robin (RR) — Preemptive (contd.)

5 **Averages :**

Avg WT = (9 + 2 + 12 + 11) ÷ 4 = 34 ÷ 4 = 8.5

Avg TAT = (14 + 5 + 20 + 17) ÷ 4 = 56 ÷ 4 = 14

Final Answer : | **Avg WT = 8.5** | **Avg TAT = 14**

Date: DD MM YY

2. Round Robin (RR) — Example B (with Idle time)

Example B : TQ = 1

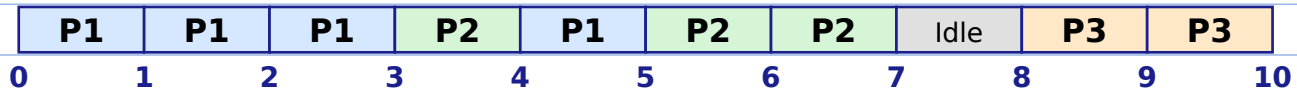
1 Given Data :

Process	AT	BT
P1	0	4
P2	3	3
P3	8	2

2 Ready Queue :

Time	Process	Remaining BT	Queue after this slot
0 - 1	P1	4 − 1 = 3	P1
1 - 2	P1	3 − 1 = 2	P1
2 - 3	P1	2 − 1 = 1	P2, P1
3 - 4	P2	3 − 1 = 2	P1, P2
4 - 5	P1	1 − 1 = 0 ✔	P2
5 - 6	P2	2 − 1 = 1	P2
6 - 7	P2	1 − 1 = 0 ✔	–
7 - 8	Idle	no process has arrived	–
8 - 9	P3	2 − 1 = 1	P3
9 - 10	P3	1 − 1 = 0 ✔	–

3 Gantt Chart :



4 Calculation : CPU stays Idle from 7 to 8

Process	AT	BT	CT	TAT = CT − AT	WT = TAT − BT
P1	0	4	5	5 − 0 = 5	5 − 4 = 1
P2	3	3	7	7 − 3 = 4	4 − 3 = 1
P3	8	2	10	10 − 8 = 2	2 − 2 = 0

5 Averages :

Avg WT = (1 + 1 + 0) ÷ 3 = 2 ÷ 3 = 0.67

Avg TAT = (5 + 4 + 2) ÷ 3 = 11 ÷ 3 = 3.67

Final Answer : | Avg WT = 0.67 | Avg TAT = 3.67

Date: DD MM YY

3. SJF (Preemptive) = SRTF — Shortest Remaining Time First

Rule : Run the process with the smallest remaining BT.

A new process with a smaller remaining BT preempts the running one.

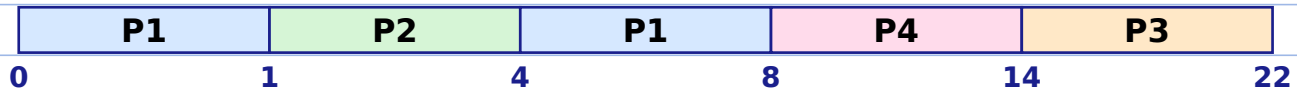
1 Given Data :

Process	AT	BT
P1	0	5
P2	1	3
P3	2	8
P4	3	6

2 Step by Step :

Time	Process	Remaining BT	Reason
0 – 1	P1	5 − 1 = 4	Only P1 has arrived
1 – 2	P2	3 − 1 = 2	P2 arrives, 3 < 4, so P1 is preempted
2 – 3	P2	2 − 1 = 1	P2 (2) is shorter than P3 (8)
3 – 4	P2	1 − 1 = 0 ✔	P2 (1) is shortest, P4 (6) has arrived
4 – 8	P1	4 − 4 = 0 ✔	Left: P1 = 4, P3 = 8, P4 = 6, so P1
8 – 14	P4	6 − 6 = 0 ✔	Left: P3 = 8, P4 = 6, so P4
14 – 22	P3	8 − 8 = 0 ✔	Only P3 is left

3 Gantt Chart :



4 Calculation : CT = end time in the Gantt chart

Process	AT	BT	CT	TAT = CT − AT	WT = TAT − BT
P1	0	5	8	8 − 0 = 8	8 − 5 = 3
P2	1	3	4	4 − 1 = 3	3 − 3 = 0
P3	2	8	22	22 − 2 = 20	20 − 8 = 12
P4	3	6	14	14 − 3 = 11	11 − 6 = 5

5 Averages :

Avg WT = (3 + 0 + 12 + 5) ÷ 4 = 20 ÷ 4 = 5

Avg TAT = (8 + 3 + 20 + 11) ÷ 4 = 42 ÷ 4 = 10.5

Final Answer : | Avg WT = 5 | Avg TAT = 10.5

Date: DD MM YY

4. Priority Scheduling — Non-preemptive

Rule : The highest priority process runs first.

Note: Here 1 = highest priority. In some questions 10 (the bigger number) is highest, so always read the convention from the question.

1 Given Data :

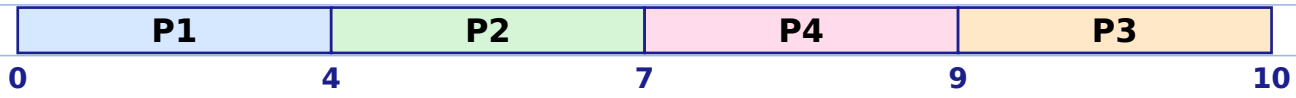
Process	AT	BT	Priority
P1	0	4	3
P2	1	3	1
P3	2	1	4
P4	3	2	2

2 Step by Step :

Time	Ready (priority)	Chosen	Runs
t = 0	P1 (3)	P1	0 – 4
t = 4	P2 (1), P3 (4), P4 (2)	P2	4 – 7
t = 7	P3 (4), P4 (2)	P4	7 – 9
t = 9	P3 (4)	P3	9 – 10

Non-preemptive: P1 is not interrupted at t = 1, 2 or 3.

3 Gantt Chart :



4 Calculation :

CT = end time in the Gantt chart

Process	AT	BT	CT	TAT = CT − AT	WT = TAT − BT
P1	0	4	4	4 − 0 = 4	4 − 4 = 0
P2	1	3	7	7 − 1 = 6	6 − 3 = 3
P3	2	1	10	10 − 2 = 8	8 − 1 = 7
P4	3	2	9	9 − 3 = 6	6 − 2 = 4

5 Averages :

Avg WT = (0 + 3 + 7 + 4) ÷ 4 = 14 ÷ 4 = 3.5

Avg TAT = (4 + 6 + 8 + 6) ÷ 4 = 24 ÷ 4 = 6

Final Answer : | **Avg WT = 3.5** | **Avg TAT = 6**